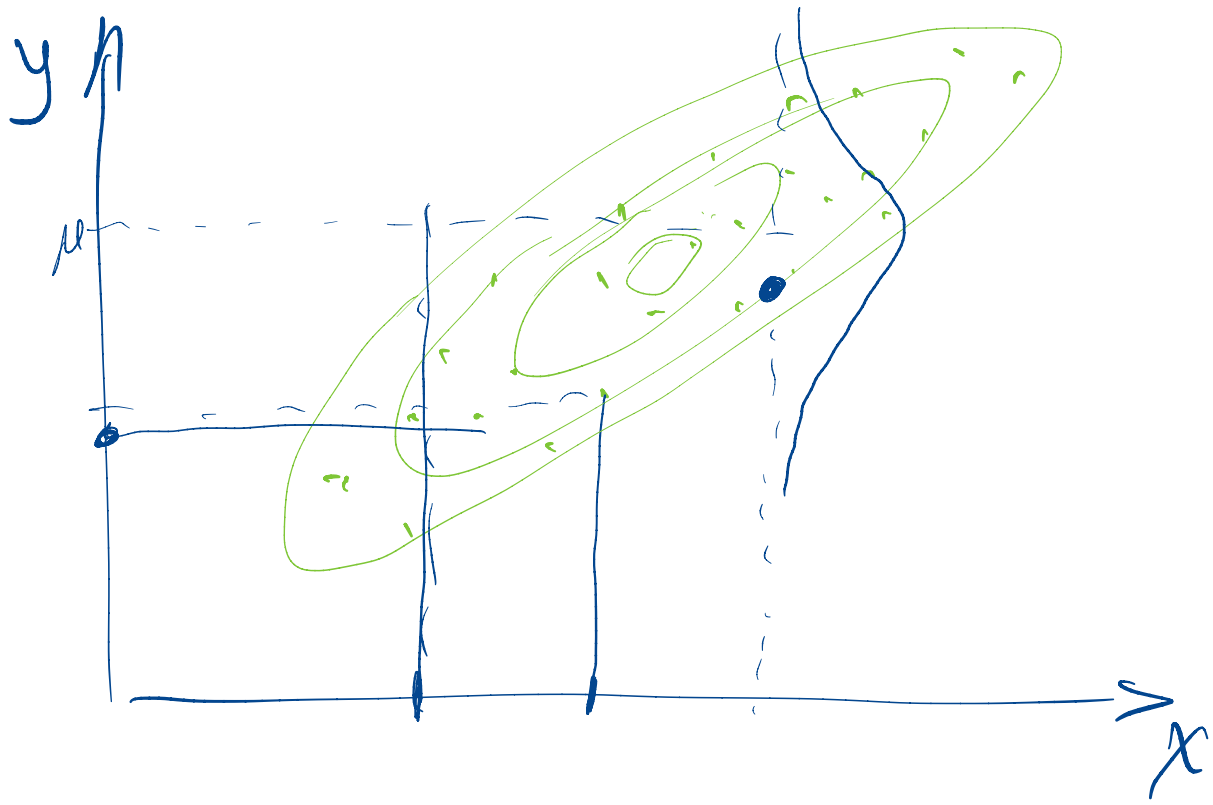


$$P(A,B) = P(A|B)\,P(B) = P(B|A)\,P(A)$$

LINE

N: $y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$

L: $\mu_i = \theta^T x_i$

I: $\{x_i, y_i\}$: i.i.d.

E: equivariance $\sigma_i^2 = \sigma^2$

$$P(\theta|\mathcal{T})\,P(\mathcal{T}) = P(\mathcal{T}|\theta)\,P(\theta)$$

$$P(\theta|\mathcal{T}) = \frac{P(\mathcal{T}|\theta)\,P(\theta)}{P(\mathcal{T})}$$

$P(\theta|\mathcal{T})$ – апостериорная в-ть θ

$P(\theta)$ – априорная в-ть θ

$P(\mathcal{T}|\theta)$ – правдоподобие выборки $\mathcal{T}$

$P(\mathcal{T})$ – evidence, в-ть выборки

$$P(\mathcal{T}|\theta) \xrightarrow[\theta]{} \max$$

$$p(y_i, x_i|\theta) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \cdot e^{-\frac{(y_i-\mu_i)^2}{2\sigma_i^2}}$$

$$L(\mathcal{T}|\theta) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_i^2}}\, e^{-\frac{(y_i-\mu_i)^2}{2\sigma_i^2}}$$

$$L(\mathcal{T}|\theta) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_i^2}} e^{-\frac{(y_i - \mu_i)^2}{2\sigma_i^2}} =$$

$$= \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_i^2}} e^{-\frac{(y_i - \theta^T x_i)^2}{2\sigma_i^2}}$$

$$\theta^* = \underset{\theta}{\operatorname{argmax}}\, L(\mathcal{T}|\theta)$$

$$\underset{\theta}{\operatorname{argmax}}\, L(\mathcal{T}|\theta) = \underset{\theta}{\operatorname{argmax}}\, \ln L(\mathcal{T}|\theta) =$$

$$= \underset{\theta}{\operatorname{argmax}} \ln \prod \frac{1}{\sqrt{\ }} e^{-\frac{(\ )^2}{2\sigma_i^2}} = \underset{\theta}{\operatorname{argmax}} \sum \ln \left(\frac{1}{\sqrt{\ }} e^{-\frac{(\ )^2}{2\sigma_i^2}} \right) =$$

$$= \underset{\theta}{\operatorname{argmax}} \left(\sum \left(\ln \frac{1}{\sqrt{2\pi\sigma_i^2}} \right) + \sum \ln e^{-\frac{(\ )^2}{2\sigma_i^2}} \right) =$$

$$= \sum \ln \frac{1}{\sqrt{2\pi\sigma^2}}$$

$$= \underset{\theta}{\operatorname{argmax}} \left(\sum_{i=1}^{N} - \frac{(y_i - \theta^T x_i)^2}{2\sigma^2} \right) =$$

$$= \underset{\theta}{\operatorname{argmax}} \sum \left(- (y_i - \theta^T x_i)^2 \right) =$$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \boxed{\sum (y_i - \theta^T x_i)^2} \qquad \hat{y}_i = \theta^T x_i$$

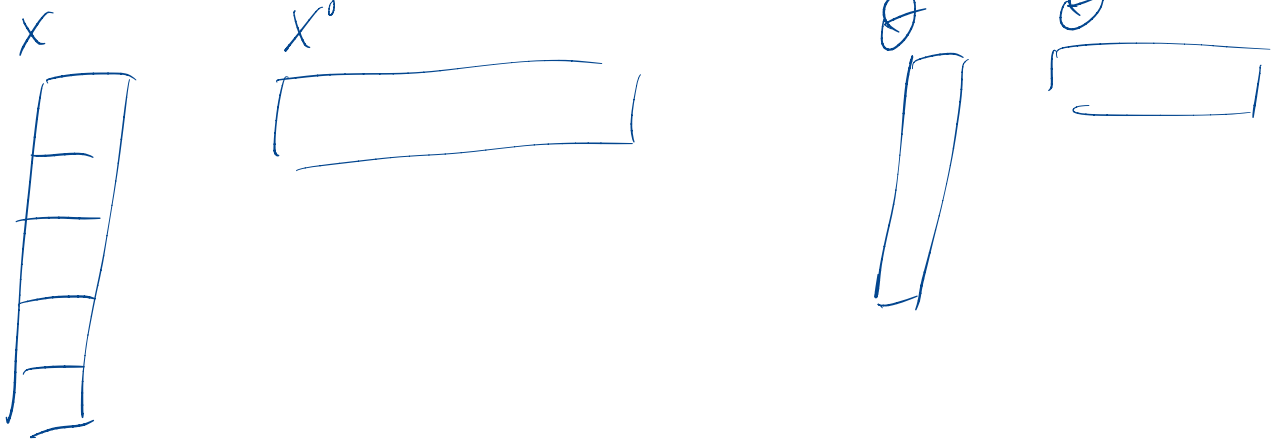
Максимизация апостериорной б-ти θ

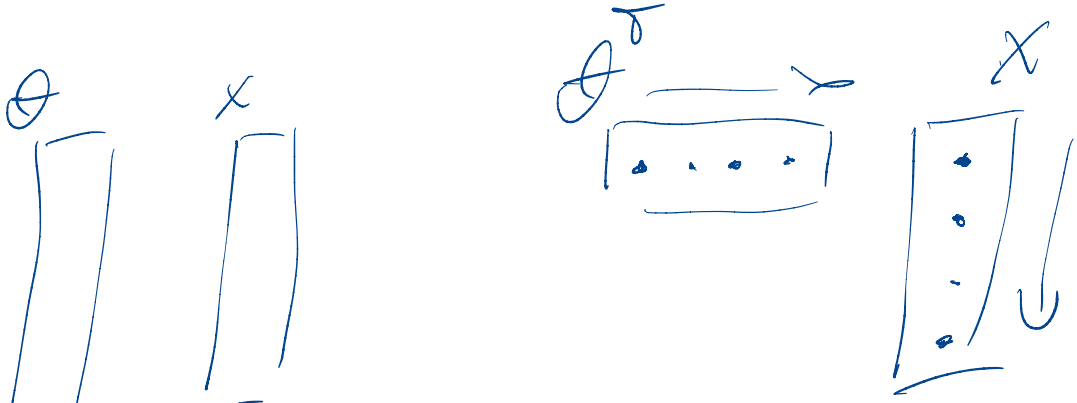
$$P(\theta | \mathfrak{D}) \to \max$$

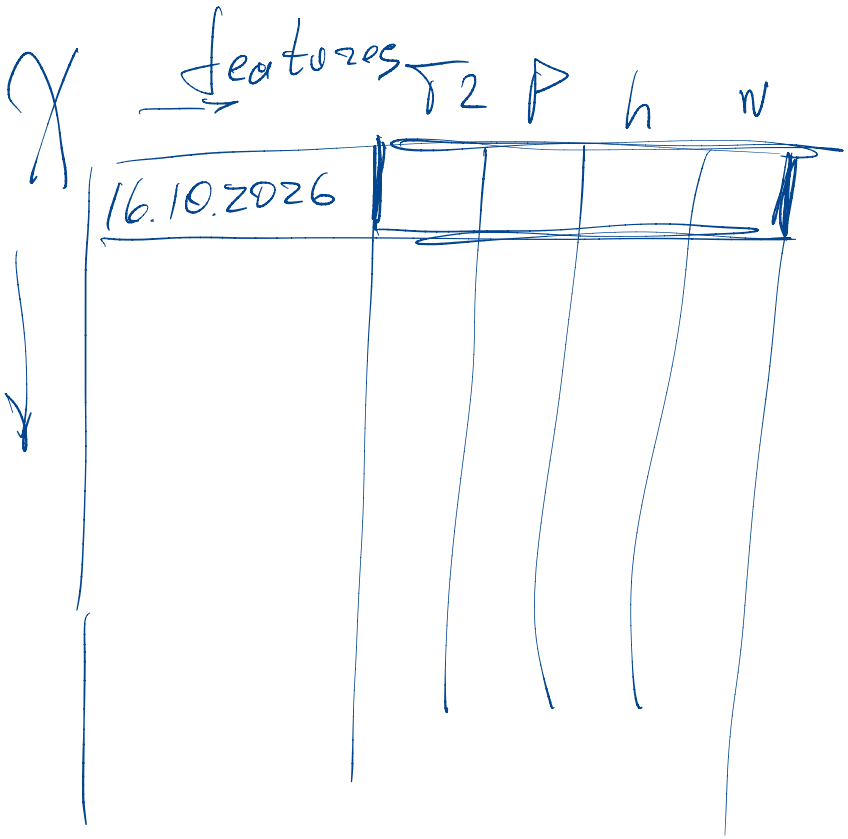
$$\theta^* = \underset{\theta}{\operatorname{argmax}} P(\theta | \mathfrak{D})$$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \boxed{\sum_{i}^{N} (y_i - \theta^T x_i)^2 + \theta^2}$$

$$\mathcal{L}(y, \theta) = \sum_{i}^{N} (y_i - \mu_i)^2$$







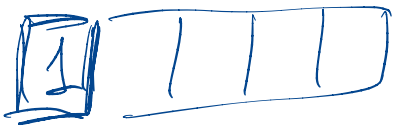
$y : N \times d$

$X : d \times N$

$X = y^T$

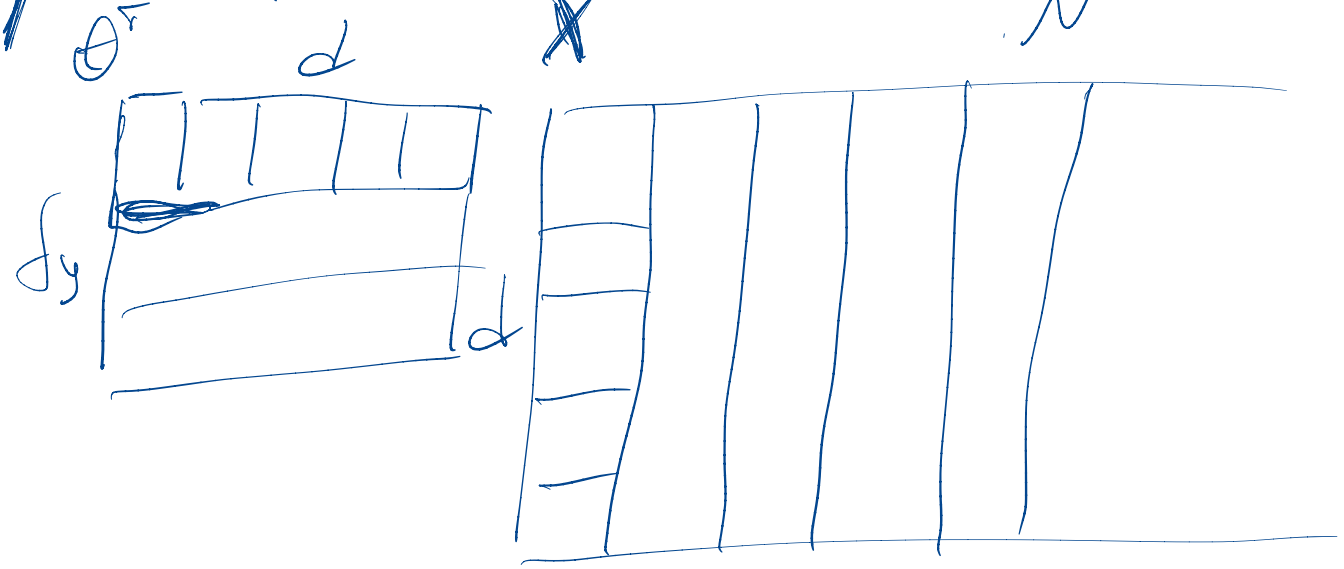
$$y = \theta_0 1 + \theta_1 x^{(1)} + \theta_2 x^{(2)} + \theta_3 x^{(3)} + \theta_4 x^{(4)}$$

$$x^{(0)} = 1$$



$$Y = \theta^T X$$

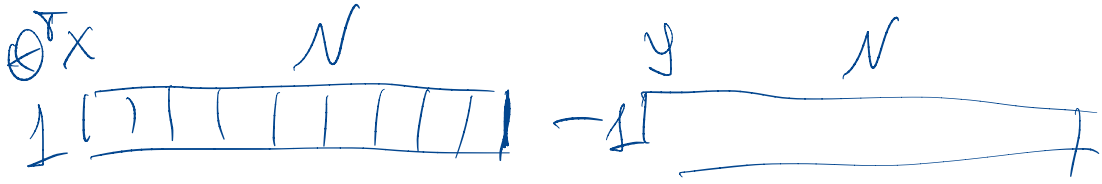
$$\theta^T$$

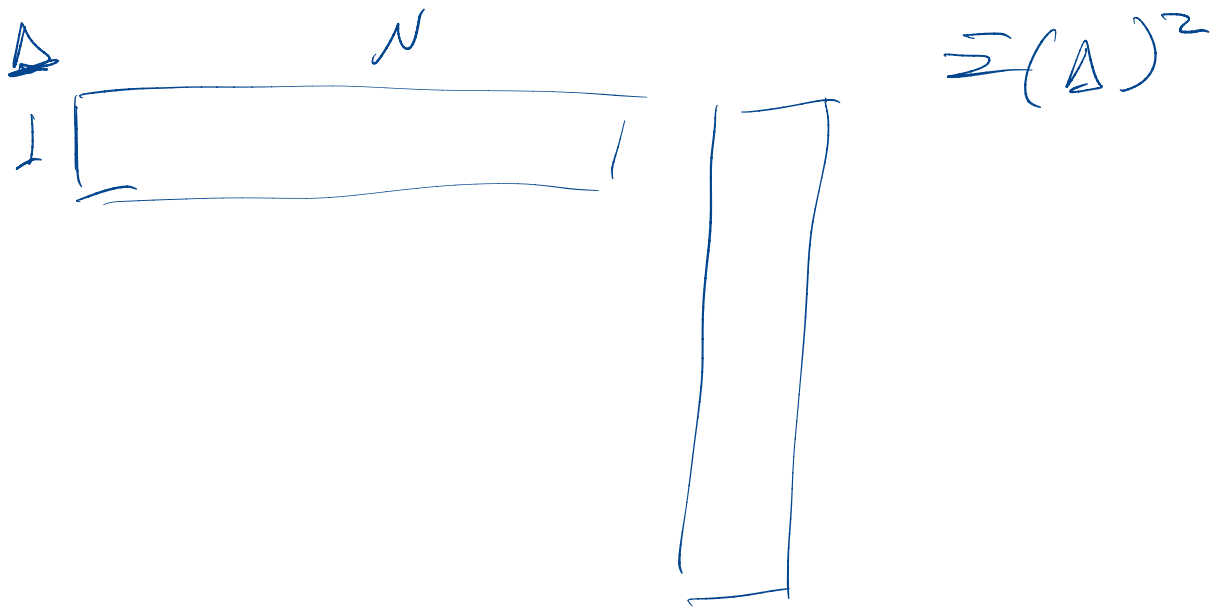


$$Y = \theta^T X$$

$$f_y \times N = f_y \times d \odot d \times N$$

$$1 \times N = 1 \times d \odot d \times N$$





$$\mathcal{L} = \Delta \Delta^T = (y - \theta^T x)(y - \theta^T x)^T =$$

$$= (\theta^T x - y)^T (\theta^T x - y) =$$

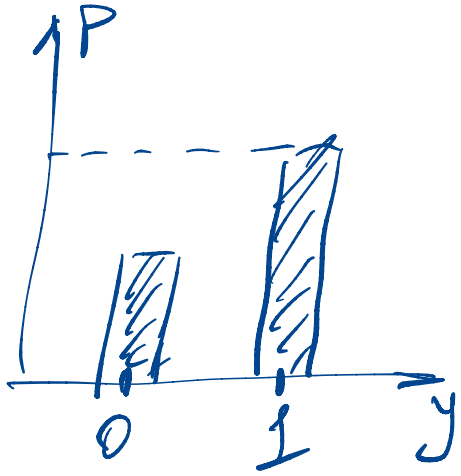
$$= y^T y - 2 (\theta^T x)^T y + (\theta^T x)^T \theta^T x$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -2 x^T y + 2 \theta^T x^T x = 0$$

$$\theta^T x^T x = x^T y$$

$$\boxed{\theta^* = (x^T y (x^T x)^{-1})^T}$$

Классификация

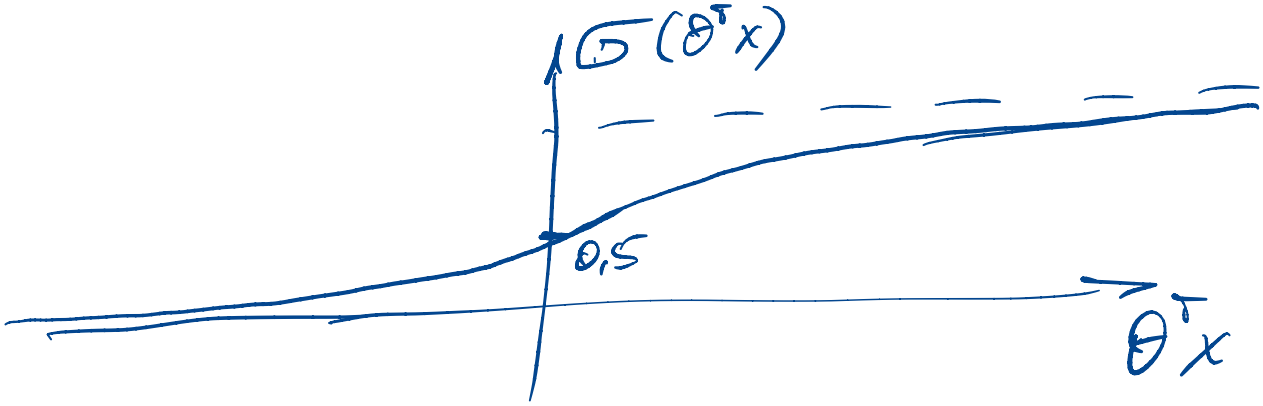


$y \sim B(p)$

$T: \{x_i; y_i\}: i.i.d.$

$p = f(\theta^T x)$

$$p = \sigma(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$



$$L_i = p_i^{y_i} \cdot (1 - p_i)^{(1 - y_i)}$$

$$L = \prod_{i=1}^{N} p_i^{y_i} (1 - p_i)^{(1 - y_i)}$$

$$\theta^* = \arg\max_{\theta} L$$

$$\ln L = \sum (\ln p_i^{y_i}) + \sum \ln (1-p_i)^{1-y_i} =$$

$$= \sum y_i \ln p_i + \sum (1-y_i) \ln (1-p_i)$$

$$p_i = \sigma(\theta^T x)$$

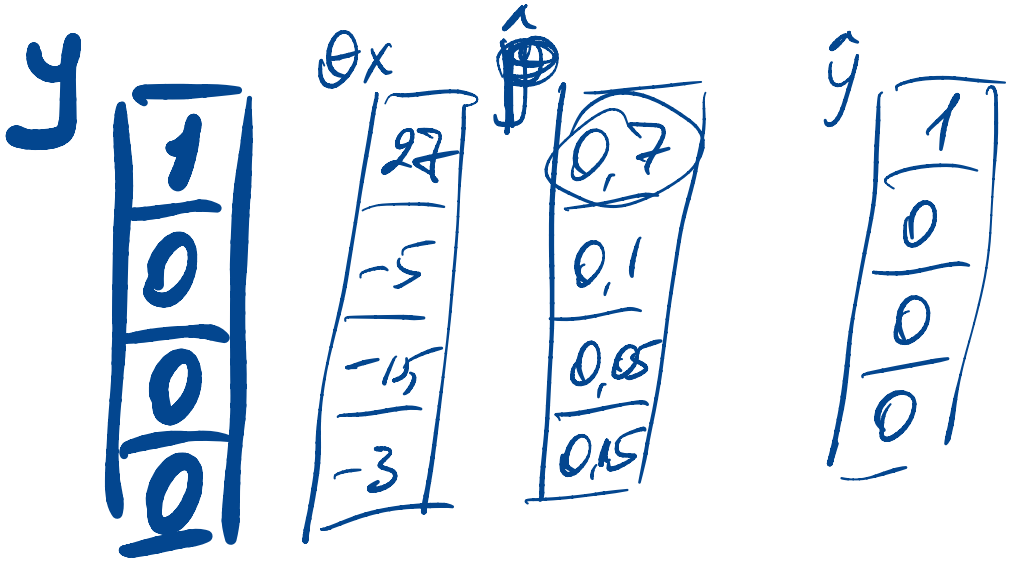
$$\mathcal{L}(T, \theta) = -\ln L = -\sum_i^N \left(y_i \ln p_i + (1-y_i) \ln (1-p_i) \right)$$

$$p_i = \sigma(\theta^T x)$$

$$\theta^* = \operatorname{argmin}(-\ln L)$$

$$+$$

nll



$$f(\theta^T x)$$

$$Softmax(z) = \frac{e^{z_j}}{\sum_{1}^{4} e^{z_j}}$$